OPEN SOURCE · PLATFORM ENGINEERING

I built a production-style Kubernetes platform that runs on a laptop.

Then I broke it on purpose, under load, and **measured how it recovered**.

```
$ make up
$ make release
$ make chaos
```

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swipe →

One command. Terraform builds the ground, Git builds the rest.

TERRAFORM	3-node kind cluster, private registry, image caches, a Git server, Argo CD
WAVE 0	Envoy Gateway (Gateway API), Kyverno, Prometheus, CloudNativePG, metrics
WAVE 1	Loki, Tempo, OpenTelemetry, routes, policies, SLO rules
WAVE 2	A Go URL shortener and its Postgres (primary + replica)

13 Argo CD Applications in three sync waves. **Nothing is installed by hand.**

no cloud account needed →

Every release goes through a gate.

01 **Build** distroless	02 **Scan** Trivy	03 **SBOM** Syft	04 **Sign** cosign
05 **Commit** Git	06 **Sync** Argo CD	07 **Admit** Kyverno	08 **Roll out** under load

0

failed requests in two releases at
50 requests a second

```
$ make unsigned
refused by Kyverno (verify-platform-images), as it
should be
```

unsigned image → no pod →

Alerts that page on what users feel, not on CPU.

99.5%
of requests answer without a 5xx

95%
of requests answer within 300 ms

- Page when the 1 h **and** 5 m burn rates pass **14.4×** (2% of the month's budget in an hour)
- Ticket at **6×** over 6 h and 30 m
- Grafana exemplars open the trace in Tempo; error logs carry the trace ID

Google SRE Workbook, ch. 5 →

Then I broke it. k6 at 50 req/s the whole time.

Release — held
live 35 s after publish · 0 of 2,299 requests failed

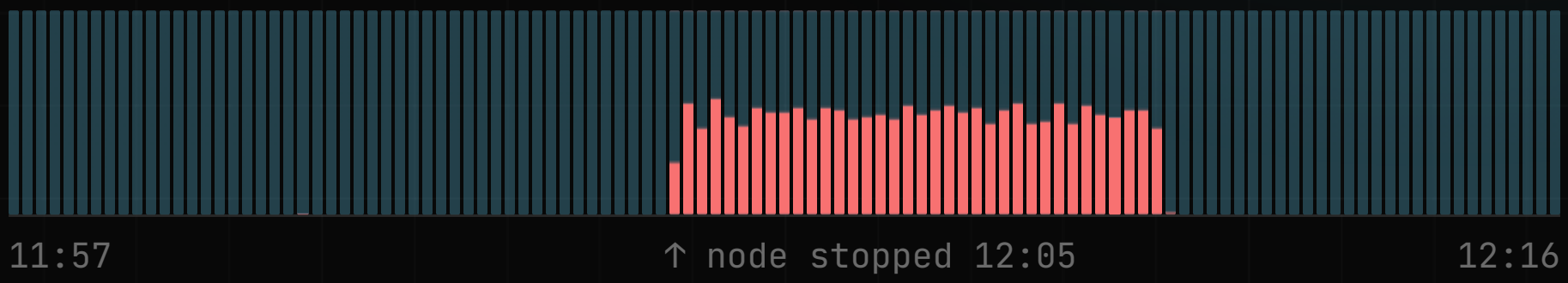
Pod kill — held
4/4 ready again in 12 s · 0 of 1,111 failed

Node loss — broke
48.4% of requests failed for 6 min 3 s

Database failover — run 2
{{NUMBER: db failover, time to a new primary}} · {{NUMBER: db failover, failed requests}}

Bad release via Git — run 2
paged after {{NUMBER: bad release, time to page}} · git revert → clean after {{NUMBER: bad release, revert to clean traffic}}

The app’s own SLO saw zero errors. Clients saw 48%.



- The gateway’s only controller was on the lost node (default 5-minute eviction)
- Kyverno’s fail-closed webhook was there too, and **blocked the evictions**
- Fixed the same day: retries + outlier detection, 2 replicas, 30 s tolerations, webhooks on CREATE/UPDATE only

After the fixes: **{{NUMBER: node loss after the fixes, run 2}}**

The AWS version is written. I haven’t applied it, and the README says so.

- EKS 1.35: spot workers, a small on-demand system group, night scale-down
- 3-AZ VPC, private nodes, one NAT (cost)
- Pod Identity for the EBS driver, a GitHub OIDC role for CI (no stored keys)
- AWS Budgets alert, S3 state with a DynamoDB lock

```
terraform validate valid
tflint 0 issues
trivy config 0 HIGH/CRITICAL
```

Run it yourself: Docker with 8 GB and one command.

github.com/Mohamed-H-Essa/k8s_platform_bootstrap	repo
docs/postmortems/2026-10-30-chaos-day.md	postmortem
mohamed-h-essa.github.io/portfolio	portfolio

I’m a mobile engineer moving into platform and DevOps work. **Questions and reviews welcome.**

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